

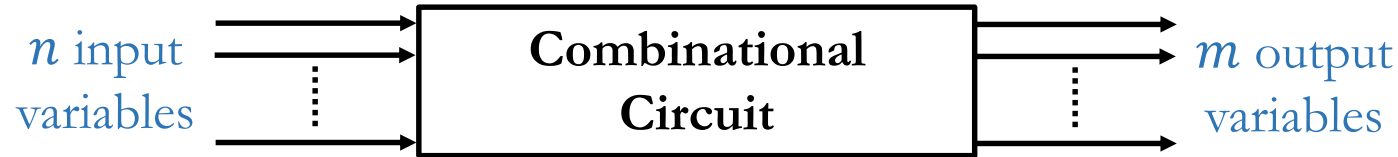
EE/CS 172/130
Digital Logic and Design

Topic 8 – Combinational Circuits

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Combinational Logic

- A **combinational circuit** is a logic circuit whose outputs at any given time depend only on the present combination of inputs, without any regard for past inputs.

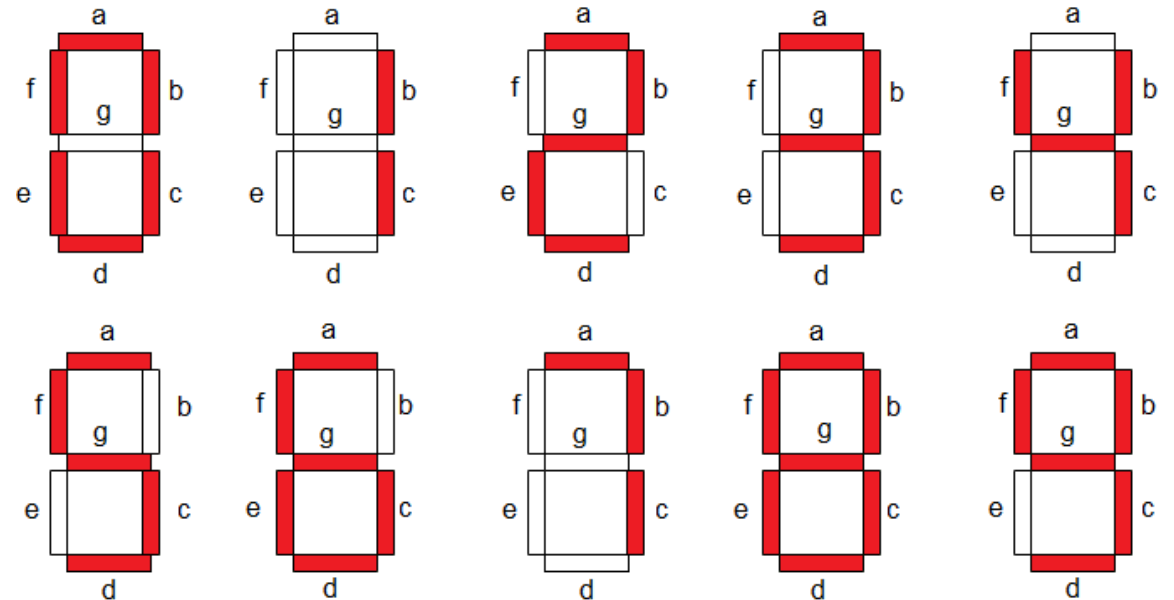
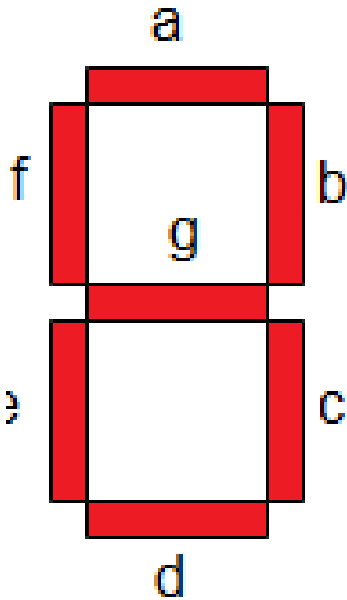


Key Characteristics

- **No Memory Elements:**
Combinational circuits do not store past input values; hence, no flip-flops or latches are used.
- **Direct Input–Output Relationship:**
The output is determined directly by the current inputs.
- **Number of Input Combinations:**
For n input variables, there are 2^n possible input combinations.
- **Unique Output for Each Input:**
For each possible input combination, there is one and only one corresponding output combination.
- **Boolean Function Representation:**
Each output variable can be expressed as a distinct Boolean function of the n input variables.

7-Segment Displays

- A **7-segment display** is an electronic display device used to represent decimal digits (0–9) and some alphabets by illuminating a combination of seven LED segments (labeled **a** through **g**) arranged in the shape of the number “8.”
- Each segment can be controlled individually to display digits 0–9 as shown below.



- A truth table or segment code is typically used to control which LEDs light up for each number.
- They are widely used in digital clocks, calculators, counters, and measurement devices.

7-Segment Displays

- There are two common configurations: Common anode (CA) and common cathode (CC).

Configuration	Description	Connection	Behavior
Common Anode (CA)	All anodes (positive terminals) of the seven LEDs are internally connected together.	Connect common pin to V_{cc} (+) . Each segment is turned ON by applying a LOW (0) signal to its cathode.	Active Low device.
Common Cathode (CC)	All cathodes (negative terminals) of the seven LEDs are internally connected together.	Connect common pin to Ground (0V) . Each segment is turned ON by applying a HIGH (1) signal to its anode.	Active High device.

TTL IC for Driving 7-Segment Displays

- TTL (Transistor-Transistor Logic) ICs are commonly used as BCD-to-7-segment decoders/drivers to convert binary-coded decimal input into the corresponding output to drive a display.
- The most commonly used TTL ICs for this purpose are:
 - IC 7447: BCD to 7-segment decoder/driver for Common Anode displays.
 - IC 7448: BCD to 7-segment decoder/driver for Common Cathode displays.
- These ICs simplify control since only a 4-bit BCD input is required, and the IC outputs signals for the correct segments.
- Current-limiting resistors are usually placed in series with each segment to protect the LEDs.

Practice Problem

- Design a BCD to 7-segment converter.
- Design a hex to 7-segment converter.